



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

AD2001-1 – FUNDAMENTALS OF MACHINE LEARNING
(AI&DS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following is a supervised learning task?

A) Clustering	B) Reinforcement Learning
C) Dimensionality Reduction	D) Classification
2. Which of the following is/are limitations of deep learning?

A) Data labeling	B) Obtain huge training datasets
C) Both A and B	D) None of the above
3. Which of the following is not a supervised machine learning algorithm?

A) K-means	B) Naïve Bayes
C) SVM for classification problems	D) Decision Tree
4. What is the key difference between supervised and unsupervised learning?

A) Supervised learning predicts labels, while unsupervised learning discovers patterns.	B) Supervised learning requires labeled data, while unsupervised learning does not.
C) Supervised learning is used for classification, while unsupervised learning is used for regression.	D) Supervised learning is always more accurate than unsupervised learning.
5. Which of the following statements is true about stochastic gradient descent?

A) It processes one training example per iteration.	B) It is not preferred, if the number of training examples is large.
C) It processes all the training examples for each iteration of gradient descent.	D) It is computationally very expensive, if the number of training examples is large.
6. What elements describe the Candidate-Elimination algorithm?

A) a) depends on the dataset.	B) just a set of candidate hypotheses.
C) just a set of instances.	D) set of instances, set of candidate hypotheses.
7. In machine learning, a model is

A) a) A physical machine that performs computations.	B) A representation of the data
C) A set of training examples	D) A type of computer algorithm
8. Baby trying to find a suitable policy to learn walking after repeated falling and getting up is an example for

A) Supervised learning	B) classification
C) Reinforcement learning	D) Unsupervised learning
9. Missing data items are with Bayes classifier.

A) ignored	B) Treated as equal compares
C) Treated as unequal compares	D) Replaced with default values
10. The Bayes rule can be used in

A) Decreasing complexity	B) Resolving queries
C) regression	D) Answering probabilistic query

11. Which optimization algorithm is commonly used to update the weights of neural networks during training?
 - A) Gradient Descent
 - B) K-Means
 - C) Apriori algorithm
 - D) Decision Trees
12. In a neural network, what are the layers between the input and output layers called?
 - A) Output layers
 - B) Hidden layers
 - C) Input layers
 - D) Feature layers
13. Which of the following statements is not true about SVM?
 - A) It is memory efficient
 - B) It can address many predictor variables
 - C) It is versatile
 - D) It doesn't require feature scaling.
14. In a logistic regression problem, there are 300 instances. 270 people voted. 30 people did not cast their votes. What is the probability of finding a person who cast one's vote?
 - A) 0.27
 - B) 0.3
 - C) 0.9
 - D) 0.1
15. How do we learn concepts from training examples?
 - A) arbitrarily
 - B) Incrementally
 - C) decremental
 - D) Non-incremental
16. Which of the following statements is false about k-Nearest Neighbor algorithm?
 - A) It cannot be used for regression
 - B) It stores all available cases and classifies new cases based on a similarity measure
 - C) It has been used in statistical estimation and pattern recognition
 - D) The input consists of the k closest training examples in the feature space
17. Which of the following is not a Decision tree algorithm?
 - A) C4.5
 - B) ID3
 - C) DBSCAN
 - D) CART
18. Which algorithm is used for finding the optimal clustering of data points?
 - A) Random Forest
 - B) K-Means clustering
 - C) SVM
 - D) Hierarchical Clustering
19. The main purpose of cross validation is to
 - A) Avoid overfitting
 - B) Avoid underfitting
 - C) Avoid noise
 - D) Increase efficiency
20. Which evaluation metric is commonly used for classification tasks when class imbalance is present?
 - A) Accuracy
 - B) Mean square Error
 - C) R-squared
 - D) F1-score

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) What is Machine Learning? Discuss its applications with merits and demerits.
- b) Differentiate Supervised Learning, Unsupervised Learning and Reinforcement learning with illustrations.
2. a) Explain the Candidate Elimination Algorithm with an example.
- b) Differentiate (i) Error and Noise (ii) Training and Testing data
- c) Apply the Find S algorithm to the following dataset (to know which seed is poisonous)

Example	Color	Toughness	Fungus	Appearance	Poisonous
1	GREEN	HARD	NO	WRINKLED	YES
2	GREEN	HARD	YES	SMOOTH	NO
3	BROWN	SOFT	NO	WRINKLED	NO
4	ORANGE	HARD	NO	WRINKLED	YES
5	GREEN	SOFT	YES	SMOOTH	YES
6	GREEN	HARD	YES	WRINKLED	YES
7	ORANGE	HARD	NO	WRINKLED	YES

Marks	BT*	CO*	PO*
8	L*2	1	2
8	L2	1	2
6	L2	2	2
4	L2	1	2
6	L3	2	3

3. a) What is gradient descent? Explain why and how gradient descent algorithm is used in Machine Learning
 b) Fit a regression line for the following data. Predict the sales for the advertisement amount of \$200.

Advertisement	Sales
\$90	\$1000
\$120	\$1300
\$150	\$1800
\$100	\$1200
\$130	\$1380

8 L2 2 3

8 L3 2 3

Unit – II

4. a) What is a classification problem? Explain how logistic regression is used in classification problems.
 b) Discuss Support Vector Machine Algorithm with an example.
5. a) The following table gives the data set about the stolen vehicles. Using Naive Bayes Classifier, classify the new data (Red, SUV, Domestic)

Color	Type	Origin	Stolen
Red	Sports	Domestic	Yes
Red	Sports	Domestic	No
Red	Sports	Domestic	Yes
Yellow	Sports	Domestic	No
Yellow	Sports	Imported	Yes
Yellow	SUV	Imported	No
Yellow	SUV	Imported	Yes
Yellow	SUV	Domestic	No
Red	SUV	Imported	No
Red	Sports	Imported	Yes

8 L2 2 3

8 L2 4 2

8 L3 3 3

8 L2 3 3

6. a) Describe K-nearest Neighbor learning Algorithm with an example.
 b) Explain (i) Locally Weighted Linear Regression (ii) Hidden Markov Model
 c) Explain on deep learning concept.

7 L2 4 2

6 L2 4 2

3 L2 4 2

Unit – III

7. a) Explain the k-Means Algorithm with an example.
 b) What is hierarchical clustering? Explain with an example.
 c) Define true error and sample error.
8. a) Discuss the learning tasks and Q learning in the context of reinforcement learning.
 b) Explain the two key difficulties that arise while estimating the accuracy of hypothesis.
 c) What is the need for resampling? Explain briefly.

6 L2 5 2

6 L2 5 2

4 L2 5 2

8 L2 5 3

4 L2 5 2

4 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
